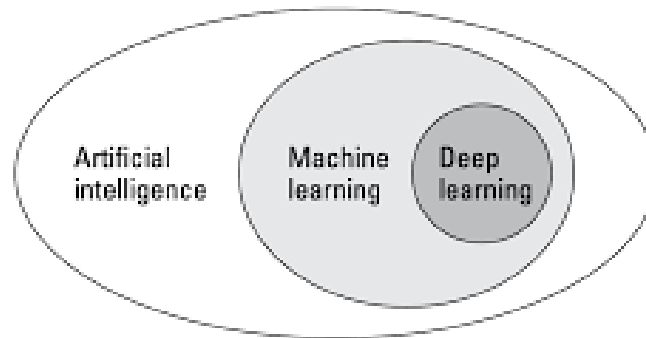


Deep Learning in Data Science



Deep learning is a powerful subset of _____ (1) learning in data science that uses artificial neural networks to model and understand complex _____ (2) in data. Inspired by the human brain, these neural networks consist of multiple _____ (3) that process data hierarchically.

Deep learning has revolutionized fields like computer vision, natural language _____ (4), and speech recognition by enabling machines to learn from vast amounts of unstructured _____ (5) such as images, text, and audio.

In data science, deep learning models are particularly useful for tasks where traditional _____ (6) struggle, such as image classification, sentiment analysis, and predictive analytics. These models require large datasets and significant computational power but often _____ (7) other methods in accuracy and efficiency.

Frameworks like TensorFlow and PyTorch have made deep learning more accessible to data scientists. Despite their complexity, these models can generalize well and are often used in real-time applications like autonomous _____ (8), recommendation systems, and medical diagnostics.

As data continues to grow in volume and variety, deep learning plays an increasingly vital _____ (9) in uncovering insights and building intelligent systems.

Choose the correct word to complete each blank.

1. Deep learning is a powerful subset of _____ learning in data science.
A) supervised
B) mechanical
C) machine
D) artificial
2. It uses neural networks to model and understand complex _____ in data.
A) decisions
B) patterns
C) rules
D) functions
3. These neural networks consist of multiple _____ that process data hierarchically.
A) blocks
B) systems
C) nodes
D) layers
4. Deep learning has transformed areas like computer vision and natural language _____.
A) development
B) understanding
C) processing
D) translation
5. It can learn from large amounts of unstructured _____ like images and audio.
A) data
B) software
C) knowledge
D) memory
6. Deep learning models are useful when traditional _____ struggle.
A) methods
B) algorithms
C) techniques
D) machines
7. These models often _____ other methods in accuracy.
A) replace
B) equal
C) outperform
D) ignore

8. Frameworks like TensorFlow and _____ help build deep learning models.
- A) Jupyter
 - B) Keras
 - C) PyTorch
 - D) NumPy
9. Deep learning is used in real-time systems like autonomous _____.
- A) navigation
 - B) driving
 - C) learning
 - D) flying
10. As data grows, deep learning plays an important _____ in building smart systems.
- A) tool
 - B) idea
 - C) role
 - D) task